MATHEMATICAL ANALYSIS 1
HOMEWORK 6

Note. This week we covered a lot of material, so this homework contains more problems than usual. For the moment it is enough to solve Problems 1–7. By the end of the semester you should also solve the Bonus problems and the Problems for later at the end of this sheet.

(1) **Indeterminate forms of algebraic type.** Compute the following limits. Explain your answers.

(a) $\lim_{x\to 0} \dfrac{x^4 - 2x^3 + 5x}{x^5 - x}$

(b) $\lim_{x\to -1} \dfrac{x+1}{\sqrt{6x^2+3}+3x}$

(c) $\lim_{x\to +\infty} \left(\sqrt{x+1} - \sqrt{x}\right)$

(d) $\lim_{x\to -\infty} \dfrac{3^x - 3^{-x}}{3^x + 3^{-x}}$

(e) $\lim_{x\to +\infty} \dfrac{x+3}{x^3 - 2x + 5}$

(f) $\lim_{x\to 0} \dfrac{\sqrt{1+\tan x} - \sqrt{1-\tan x}}{\sin x}$

(g) Determine the behavior of $f(x) = \frac{x^3 - x^2 + 3}{x^2 + 3x + 2}$ as $x \to +\infty$ and as $x \to -\infty$. (In Homework 5, Problem 9, you studied this function near $x = -2$ and $x = -1$.)

(2) **The Substitution Theorem.** Whenever you use the Substitution Theorem (Theorem 6.12 of the lecture notes), say which functions play the roles of f and g, which of the three cases of the theorem you use, and check its hypotheses.

(a) Compute $\lim_{x\to+\infty} x \sin\frac{1}{x}$ and $\lim_{x\to 0} x \sin\frac{1}{x}$. Why do the two limits require different tools?

(b) Deduce the value of $\lim_{x\to+\infty} \left(5 + 2x \sin\dfrac{1}{x}\right)$.

(c) Compute $\lim_{x\to 1} \dfrac{\cos\left(\frac{\pi}{2}x\right)}{1-x}$. *Hint: take $y = 1 - x$.*

(d) Compute the following limits (one-sided where indicated):

$$\lim_{x\to 0} e^{-1/x^2}, \qquad \lim_{x\to 0^+} e^{1/x}, \qquad \lim_{x\to 0^-} e^{1/x},$$

$$\lim_{x\to 0^+} \ln(\sin x), \qquad \lim_{x\to 0^+} \arctan\frac{1}{x}, \qquad \lim_{x\to 0^-} \arctan\frac{1}{x}.$$

(3) **Limits of sequences.** Determine how the following sequences $\{a_n\}_{n\in\mathbb{N}_+}$ behave as $n \to \infty$ (converge, diverge to $\pm\infty$, or are indeterminate). Explain your answers.

(a) $a_n = n - \sqrt{n}$

(b) $a_n = \frac{(2n)!}{n!}$

(c) $a_n = \frac{(2n)!}{(n!)^2}$

(d) $a_n = \binom{n}{3}\frac{6}{n^3}$

(e) $a_n = 2^n \sin\left(2^{-n}\pi\right)$

(f) $a_n = n\cos\left(\frac{n+1}{n} \cdot \frac{\pi}{2}\right)$

(4) **Fundamental limits.** Compute the following limits. Explain your answers.

(a) $\lim_{x\to 0} \dfrac{x - \tan x}{x^2}$ *(hint: Squeeze Theorem, and the inequality $\sin x < x < \tan x$ for $0 < x < \frac{\pi}{2}$ from Example 5.13 of the lecture notes)*

(b) $\lim_{x\to e} \dfrac{\ln x - 1}{x - e}$ *(hint: take $y = x - e$)*

(c) $\lim_{x\to 0^+} \dfrac{2^{2x} - 1}{2x}$

(d) $\lim_{x\to 1} \dfrac{\ln x}{e^x - e}$ *(hint: take $y = x - 1$)*

(e) $\lim_{x\to 0} \dfrac{e^x - e^{-x}}{\sin x}$

(f) Determine the domain of $f(x) = \sqrt[3]{x}\, e^{-x^2}$ and the behavior of f at the end-points of its domain. *(Hint: you may use the fact that* $\lim_{x\to+\infty}\left(\frac{1}{3}\ln x - x^2\right) = -\infty$*.)*

(5) **Two applications.**

(a) *Radioactive decay.* A radioactive sample contains $N(t) = N_0\, 2^{-t/T}$ atoms at time $t \geq 0$, where $N_0 > 0$ is the initial number of atoms and $T > 0$ is the half-life.

(i) Compute the initial rate of decay $\lim\limits_{h\to 0^+} \dfrac{N(h) - N_0}{h}$.

(ii) Show that $\lim\limits_{n\to\infty} n\left(2^{1/n} - 1\right) = \ln 2$.

(b) *Charging a capacitor.* In an RC circuit (resistance $R > 0$, capacitance $C > 0$) connected at time $t = 0$ to a battery of voltage $V_0 > 0$, the voltage across the capacitor is $V(t) = V_0\left(1 - e^{-t/(RC)}\right)$ for $t \geq 0$.

(i) Compute the final voltage $\lim_{t\to+\infty} V(t)$.

(ii) Compute the initial rate of charging $\lim\limits_{t\to 0^+} \dfrac{V(t)}{t}$.

(iii) The number $\tau = RC$ is called the *time constant* of the circuit. What fraction of the final voltage has been reached at time $t = \tau$?

(6) **Understanding Landau symbols.** Justify each answer using the definitions of the Landau symbols.

(a) Consider the function $4 + \frac{\sin x}{x}$ as $x \to \pm\infty$.

(i) Show that $\frac{\sin x}{x} = o(x)$ as $x \to \pm\infty$, and deduce that $4 + \frac{\sin x}{x} = 4 + o(x)$ as $x \to \pm\infty$.

(ii) Show that also $\frac{\sin x}{x} = o(1)$ and $\frac{\sin x}{x} = \mathcal{O}\left(\frac{1}{x}\right)$ as $x \to \pm\infty$.

(iii) Which of the three statements $4+\frac{\sin x}{x} = 4+o(x)$, $4+\frac{\sin x}{x} = 4+o(1)$, $4+\frac{\sin x}{x} = 4+\mathcal{O}\left(\frac{1}{x}\right)$ gives the most information? Give an example of a function f with $f(x) = 4 + o(x)$ as $x \to +\infty$ which does not tend to 4.

(b) True or false, as $x \to +\infty$?

$$x^2 + 3x = x^2 + o(x^2), \quad x^2 + 3x = \mathcal{O}(x), \quad \sqrt{x} = o(x), \quad x + \sin x \sim x, \quad x + \sin x = x + \mathcal{O}(1), \quad x^2 = o(x^3).$$

(c) True or false, as $x \to 0$? (Compare with (b): near 0 the larger powers are the negligible ones.)

$$x^2 + x^3 = o(x), \quad x^3 = o(x^2), \quad x = o(x^2), \quad \sin x = x + o(x),$$
$$1 - \cos x = o(x), \quad e^x = 1 + x + o(x), \quad \ln(1 + x) = x + o(x).$$

(d) *Warnings.*

(i) Show that $x^2 + x \sim x^2$ as $x \to +\infty$, but that $(x^2 + x) - x^2$ does not tend to 0. So $f \sim g$ does not mean that $f - g \to 0$.

(ii) As $x \to 0$ we have $x^2 = o(x)$ and $2x^2 = o(x)$, but $x^2 \neq 2x^2$. Explain why "$o(g) - o(g) = 0$" is wrong.

(e) *Relativistic kinetic energy.* A body of mass m moving with speed v has kinetic energy $(\gamma - 1)mc^2$, where c is the speed of light, $\beta = \frac{v}{c}$ and $\gamma = \left(1 - \beta^2\right)^{-1/2}$. Using $\lim_{x\to 0} \frac{(1+x)^\alpha - 1}{x} = \alpha$ and the Substitution Theorem, show that $\gamma = 1 + \frac{\beta^2}{2} + o(\beta^2)$ as $\beta \to 0$. Conclude that $(\gamma - 1)mc^2 = \frac{1}{2}mv^2 + o(v^2)$ as $v \to 0$: at low speeds we recover the classical kinetic energy.

(7) **Computing with little o.** Use Proposition 8.1 and Corollary 8.2 of the lecture notes and the comparison of powers, and say at each step which rule you use. Recall that such identities are read from left to right.

(a) Show that, as $x \to 0$,

$$\left(x + 3x^2 + o(x^2)\right)\left(2 - x + o(x)\right) = 2x + 5x^2 + o(x^2).$$

Can you also determine the coefficient of x^3 in this product? Why or why not?

(b) *Thermal expansion.* A metal cube has side length $L_0 > 0$ at a reference temperature θ_0. When the temperature changes to $\theta_0 + T$, its side length is $L(T)$. Here T may be positive (heating) or negative (cooling). Experiments show that

$$L(T) = L_0\left(1 + \alpha T + o(T)\right) \qquad \text{as } T \to 0,$$

where $\alpha > 0$ is the coefficient of linear expansion of the metal (for steel, $\alpha \approx 1.2 \cdot 10^{-5}$ K^{-1}). Show that the area of a face is $A(T) = L_0^2\,(1 + 2\alpha T + o(T))$ and the volume is $V(T) = L_0^3\,(1 + 3\alpha T + o(T))$ as $T \to 0$. In words: for small temperature changes, the relative change of the area is about twice, and that of the volume about three times, the relative change of the length.

(c) *Electrical power.* As $t \to 0$, the voltage and the current in a circuit are $V(t) = V_0\,(1 + 2t + o(t))$ and $I(t) = I_0\,(1 - t + o(t))$. Show that the power $P = VI$ satisfies $P(t) = V_0 I_0\,(1 + t + o(t))$ as $t \to 0$. In words: for small t, the relative change of the power is approximately the sum of the relative changes of the voltage and of the current.

(d) *Precision is limited by the worst term.* Show that, as $x \to 0$, $(2 + o(1))\,\big(x - x^3 + o(x^3)\big) = 2x + o(x)$. Explain why we cannot conclude that the product equals $2x - 2x^3 + o(x^3)$.

(e) *Subtracting does not cancel the o's.* Show that as $x \to 0$ $\big(x + 2x^2 + o(x^2)\big) - \big(x + 2x^2 + o(x^3)\big) = o(x^2)$. Why is the answer not 0?

(f) *At infinity the roles are reversed.* Show that, as $x \to +\infty$,

$$\big(x^2 + x + o(x)\big)\left(1 + \frac{1}{x} + o\left(\frac{1}{x}\right)\right) = x^2 + 2x + o(x).$$

Hint: as $x \to +\infty$, every constant is $o(x)$, and $o(1) = o(x)$.

Bonus problems

(B1) **The Ratio Test for sequences.** In this problem we prove Theorem 6.16 of the lecture notes (you are guided in the steps below). Let $\{a_n\}_{n\in\mathbb{N}}$ be a sequence with $a_n > 0$ for all large n, and suppose that the limit $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = q$ exists (finite or $+\infty$). You may use that $\lim_{k\to\infty} r^k = 0$ if $0 < r < 1$, and $\lim_{k\to\infty} r^k = +\infty$ if $r > 1$.

(a) State Theorem 6.16.

(b) Suppose that $q < 1$. Explain why $q \geq 0$ (*hint: Theorem 6.15, item 4*), and choose a number r with $q < r < 1$.

(c) Use the definition of the limit with $\varepsilon = r - q$ to show that there exists $N \in \mathbb{N}$ such that $a_n > 0$ and $a_{n+1} < r\, a_n$ for all $n > N$.

(d) Let $M = N + 1$. Prove by induction on k that $0 < a_{M+k} \leq r^k a_M$ for all $k \in \mathbb{N}$.

(e) Conclude that $\lim_{n\to\infty} a_n = 0$ (*hint: Theorem 6.15, item 5(a)*).

(f) Now suppose that $q > 1$ (including $q = +\infty$). Choose r with $1 < r < q$ (for instance $r = 2$ if $q = +\infty$), show as in (c) and (d) that $a_{M+k} \geq r^k a_M$ for all $k \in \mathbb{N}$, and conclude that $\lim_{n\to\infty} a_n = +\infty$ (*hint: Theorem 6.15, item 5(b)*).

(g) Give three sequences with $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = 1$ whose limits are 0, 1 and $+\infty$, respectively. Why does this show that no conclusion is possible when $q = 1$?

(h) Use the theorem to compute $\displaystyle\lim_{n\to\infty} \frac{n^{10}}{1.1^n}$ and $\displaystyle\lim_{n\to\infty} \frac{100^n}{n!}$. (An algorithm whose running time grows like 1.1^n is therefore eventually slower than one whose running time grows like n^{10}, even though for $n = 100$ we have $1.1^{100} \approx 1.4 \cdot 10^4$ while $100^{10} = 10^{20}$.)

(B2) **Proofs with Landau symbols.** Throughout this problem $f, g, h : \mathbb{R} \to \mathbb{R}$, $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$, every function that appears in a denominator is nonzero on a deleted neighborhood of x_0, and all statements are as $x \to x_0$.

(a) (i) Prove that if $f \sim g$ then $f \asymp g$.

(ii) Prove that if $f \asymp g$, with $\lim_{x\to x_0} \frac{f(x)}{g(x)} = \ell$, then $f \sim \ell g$.

(iii) Give an example showing that $f = \mathcal{O}(g)$ does not imply $f \asymp g$.

(b) Prove that $f \sim g$ if and only if $f = g + o(g)$. *Hint: $f = g + o(g)$ means that $f - g = o(g)$.*

(c) (i) Prove that if $f \sim g$ then $g \sim f$. *Hint: first show, using Theorem 6.2 of the lecture notes, that $f(x) \neq 0$ on a deleted neighborhood of x_0.*

(ii) Prove that if $f \sim g$ and $g \sim h$ then $f \sim h$.

(iii) Prove that if $f = o(g)$ and $g = \mathcal{O}(h)$ then $f = o(h)$.

(d) Let $x_0 \in \mathbb{R}$ and let f be defined on a neighborhood of x_0. Prove that f is continuous at x_0 if and only if $f(x) = f(x_0) + o(1)$ as $x \to x_0$.

(L1) **Series.**

(a) *Geometric series.*

(i) Let $r \in \mathbb{R}$, $r \neq 1$. Show that the partial sums of the geometric series satisfy

$$s_n = \sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r}.$$

(ii) Using a geometric series, write the number $2.3\overline{17} = 2.3171717\cdots$ as a fraction.

(iii) *Bouncing ball.* A ball is dropped from a height of 10 m. Each time it hits the ground, it bounces back up to 60% of the height from which it fell. Find the total distance traveled by the ball. Falling from height h takes $\sqrt{2h/g}$ seconds, and so does rising to height h. Show that the ball stops bouncing after a finite time, and compute this time (take $g = 9.81$ m/s^2).

(iv) *Drug dosing.* A patient takes 100 mg of a drug every 24 hours. After 24 hours, 25% of the amount of drug that was in the body remains. Find the amount of drug in the body just after the n-th dose, and its limit as $n \to \infty$.

(b) *Cauchy Condensation Test.* Let $q > 0$. Use Theorem 7.1 of the lecture notes to show that $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^q}$ converges if and only if $q > 1$.

(c) *The necessary condition.* Show that $\displaystyle\sum_{k=2}^{\infty} (-1)^k \sqrt{\frac{k^3+3}{2k^3-5}}$ does not converge. Why can't Leibniz's Test be applied?

(d) *Comparison and Asymptotic Comparison Tests.* Using one of these tests, decide whether the following series converge:

$$\text{(i)} \sum_{k=1}^{\infty} \frac{1}{2^k - 1} \qquad \text{(ii)} \sum_{k=1}^{\infty} \frac{k+3}{\sqrt[3]{k^9 + k^2}}$$

(e) *Ratio and Root Tests.* Using one of these tests, decide whether the following series converge:

$$\text{(i)} \sum_{n=1}^{\infty} \frac{n!}{n^n} \qquad \text{(ii)} \sum_{n=1}^{\infty} \frac{n^n}{n!} \qquad \text{(iii)} \sum_{n=1}^{\infty} \frac{2^n\, n!}{n^n} \text{ and } \sum_{n=1}^{\infty} \frac{3^n\, n!}{n^n}$$

$$\text{(iv)} \sum_{n=1}^{\infty} \frac{n^3}{3^n} \qquad \text{(v)} \sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$$

(f) *Leibniz's Alternating Series Test.* Show that each of the following series converges. Does it converge absolutely or only conditionally?

$$\text{(i)} \sum_{k=1}^{\infty} \sin\left(k\pi + \frac{1}{k}\right) \qquad \text{(ii)} \sum_{k=1}^{\infty} (-1)^{k+1} \frac{k^2}{k^3+1} \qquad \text{(iii)} \sum_{k=1}^{\infty} (-1)^k \ln\left(1 + \frac{1}{k}\right)$$

Hint for (i): $\sin(k\pi + t) = (-1)^k \sin t$.

(g) *Absolute convergence.*

(i) Does $\displaystyle\sum_{k=1}^{\infty} \frac{(-1)^{k+1} 5^{k-1}}{(k+1)^2\, 4^{k+2}}$ converge absolutely? Does it converge?

(ii) Does $\displaystyle\sum_{k=1}^{\infty} \frac{\cos 3k}{k^3}$ converge absolutely?

(L2) **The harmonic series.** For $n \in \mathbb{N}_+$ let $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$ be the n-th partial sum of the harmonic series.

(a) *Oresme's proof that the harmonic series diverges.*

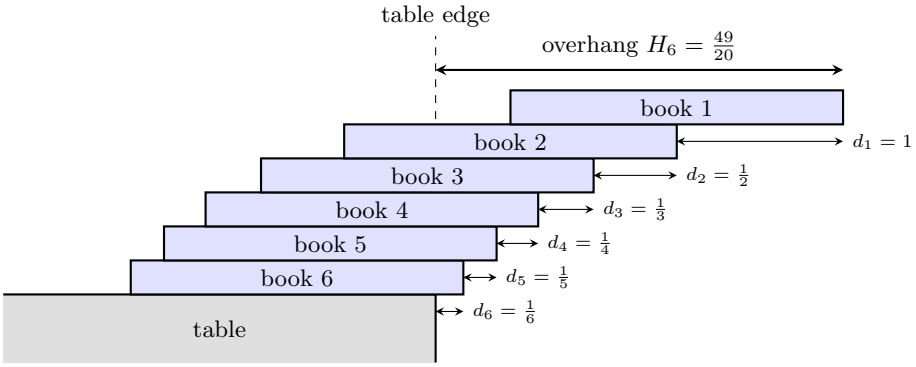
(i) Show that for every $k \in \mathbb{N}_+$, the 2^{k-1} numbers $\frac{1}{2^{k-1}+1}, \frac{1}{2^{k-1}+2}, \ldots, \frac{1}{2^k}$ add up to at least $\frac{1}{2}$.

(ii) Deduce that $H_{2^m} \geq 1 + \frac{m}{2}$ for every $m \in \mathbb{N}$.

(iii) Conclude that $\lim_{n\to\infty} H_n = +\infty$.

(b) The harmonic series diverges very slowly: using (a), find some n for which $H_n \geq 10$.

(c) *Book stacking.* We stack n identical books, each of length 2, at the edge of a table, as in the figure below: book 1 is on top and book n lies on the table. Let d_k be the distance by which the right edge of book k sticks out beyond the right edge of the book below it (for $k = n$: beyond the edge of the table). We use three facts from physics: the center of mass of a single book is at its midpoint; the center of mass of several identical books is the average of their centers of mass; and a stack resting on a support does not tip over as long as its center of mass is not beyond the edge of the support.



(i) Explain why the largest possible value of d_1 is 1, and why in that case the center of mass of book 1 lies exactly above the right edge of book 2.

(ii) Suppose that the center of mass of the top k books lies exactly above the right edge of book $k+1$, and put the origin of the x-axis at this edge. Where is the center of mass of book $k+1$?

(iii) Show that the center of mass of the top $k+1$ books is at $x = -\frac{1}{k+1}$.

(iv) Deduce that the largest possible value of d_{k+1} is $\frac{1}{k+1}$, and that in that case the center of mass of the top $k+1$ books lies exactly above the right edge of book $k+2$ (or of the table, if $k+1 = n$).

(v) Conclude by induction that with $d_k = \frac{1}{k}$ for $k = 1, \ldots, n$ the right edge of the top book is at distance H_n beyond the edge of the table.

(vi) With these values every part of the stack is exactly balanced, so in practice the overhang can only be made arbitrarily close to H_n. Using (a), find a number of books that allows an overhang larger than 3. Then check with a calculator that 11 books suffice, but 10 do not.

(vii) Can the top book be arbitrarily far beyond the edge of the table?

Homework 6 Solutions

(1) (a) Numerator and denominator both tend to 0. For $x \neq 0$ we can cancel the factor x:

$$\frac{x^4 - 2x^3 + 5x}{x^5 - x} = \frac{x(x^3 - 2x^2 + 5)}{x(x^4 - 1)} = \frac{x^3 - 2x^2 + 5}{x^4 - 1} \xrightarrow{x \to 0} \frac{5}{-1} = \mathbf{-5},$$

since the last expression is a rational function whose denominator does not vanish at 0, hence continuous at 0 (Corollary 6.7).

(b) At $x = -1$ the denominator is $\sqrt{9} - 3 = 0$, so this is a form $\frac{0}{0}$. Multiplying numerator and denominator by $\sqrt{6x^2+3} - 3x$ (which tends to $6 \neq 0$):

$$\frac{x+1}{\sqrt{6x^2+3}+3x} = \frac{(x+1)\left(\sqrt{6x^2+3}-3x\right)}{(6x^2+3)-9x^2} = \frac{(x+1)\left(\sqrt{6x^2+3}-3x\right)}{3(1-x)(1+x)} = \frac{\sqrt{6x^2+3}-3x}{3(1-x)}$$

for $x \neq \pm 1$ near -1. The last expression is continuous at -1 (Corollary 6.13 for the square root), so the limit is $\frac{3+3}{3\cdot 2} = \mathbf{1}$.

(c) This is a form $\infty - \infty$. Rationalizing,

$$\sqrt{x+1} - \sqrt{x} = \frac{(x+1)-x}{\sqrt{x+1}+\sqrt{x}} = \frac{1}{\sqrt{x+1}+\sqrt{x}} \xrightarrow{x \to +\infty} \mathbf{0},$$

since the denominator tends to $+\infty$.

(d) Multiplying numerator and denominator by $3^x > 0$,

$$\frac{3^x - 3^{-x}}{3^x + 3^{-x}} = \frac{3^{2x} - 1}{3^{2x} + 1}.$$

As $x \to -\infty$ we have $2x \to -\infty$ and $\lim_{y \to -\infty} 3^y = 0$, so $3^{2x} \to 0$ (Theorem 6.12, case 3, with $f(x) = 2x$, $g(y) = 3^y$). Hence the limit is $\frac{0-1}{0+1} = \mathbf{-1}$.

(e) Dividing numerator and denominator by x^3:

$$\frac{x+3}{x^3 - 2x + 5} = \frac{\frac{1}{x^2} + \frac{3}{x^3}}{1 - \frac{2}{x^2} + \frac{5}{x^3}} \xrightarrow{x \to +\infty} \frac{0}{1} = \mathbf{0}.$$

(f) Both numerator and denominator tend to 0. Multiplying by $\sqrt{1+\tan x} + \sqrt{1-\tan x}$, for $x \neq 0$ near 0:

$$\frac{\sqrt{1+\tan x} - \sqrt{1-\tan x}}{\sin x} = \frac{(1+\tan x)-(1-\tan x)}{\sin x\left(\sqrt{1+\tan x}+\sqrt{1-\tan x}\right)} = \frac{2}{\cos x\left(\sqrt{1+\tan x}+\sqrt{1-\tan x}\right)},$$

using $\frac{\tan x}{\sin x} = \frac{1}{\cos x}$. The last expression is continuous at 0 (Corollaries 6.6 and 6.13), so the limit is $\frac{2}{1\cdot(1+1)} = \mathbf{1}$.

(g) For $x \neq 0$ (and $x \neq -1, -2$),

$$f(x) = \frac{x^3\left(1 - \frac{1}{x} + \frac{3}{x^3}\right)}{x^2\left(1 + \frac{3}{x} + \frac{2}{x^2}\right)} = x \cdot \frac{1 - \frac{1}{x} + \frac{3}{x^3}}{1 + \frac{3}{x} + \frac{2}{x^2}}.$$

The fraction tends to 1 as $x \to \pm\infty$, so by the algebra of limits $\lim_{x \to +\infty} f(x) = \mathbf{+\infty}$ and $\lim_{x \to -\infty} f(x) = \mathbf{-\infty}$.

(2) (a) *As $x \to +\infty$* (a form $\infty \cdot 0$): write $x \sin\frac{1}{x} = g(f(x))$ with $f(x) = \frac{1}{x}$ and $g(y) = \frac{\sin y}{y}$. Then $\lim_{x \to +\infty} f(x) = 0 = \ell \in \mathbb{R}$, $f(x) \neq 0$ for all $x > 0$, and $\lim_{y \to 0} g(y) = 1$ by Example 5.13. By Theorem 6.12, **case 2**,

$$\lim_{x \to +\infty} x \sin\frac{1}{x} = \lim_{y \to 0} \frac{\sin y}{y} = \mathbf{1}.$$

As $x \to 0$: here $x \to 0$ and $\left|\sin\frac{1}{x}\right| \leq 1$ for all $x \neq 0$, so $\sin\frac{1}{x}$ is bounded and, by the Squeeze to 0 Theorem (Corollary 6.10), $\lim_{x \to 0} x \sin\frac{1}{x} = \mathbf{0}$.

The tools differ because the behavior differs: at $+\infty$ the factor $\sin\frac{1}{x}$ tends to 0 and we have a genuine indeterminate form, which the substitution $y = \frac{1}{x}$ turns into the fundamental limit

$\frac{\sin y}{y}$; at 0 the factor $\sin\frac{1}{x}$ has *no* limit (Homework 5, Problem 5), so no substitution can help, but it is bounded, and the factor x tends to 0.

(b) By (a) and the algebra of limits, $\lim_{x\to+\infty}\left(5+2x\sin\frac{1}{x}\right)=5+2\cdot 1=\mathbf{7}$.

(c) Let $y=1-x$, i.e. $f(x)=1-x$; then $f(x)\to 0$ as $x\to 1$ and $f(x)\neq 0$ for $x\neq 1$. Since

$$\cos\left(\frac{\pi}{2}x\right)=\cos\left(\frac{\pi}{2}-\frac{\pi}{2}y\right)=\sin\left(\frac{\pi}{2}y\right),$$

we have $\frac{\cos(\frac{\pi}{2}x)}{1-x}=g(f(x))$ with $g(y)=\frac{\sin(\frac{\pi}{2}y)}{y}$. To compute $\lim_{y\to 0}g(y)$ we substitute once more: $g(y)=\frac{\pi}{2}\cdot\frac{\sin t}{t}$ with $t=\frac{\pi}{2}y$, which tends to 0 and is $\neq 0$ for $y\neq 0$; by case 2 of Theorem 6.12 and Example 5.13, $\lim_{y\to 0}g(y)=\frac{\pi}{2}$. By case 2 again,

$$\lim_{x\to 1}\frac{\cos\left(\frac{\pi}{2}x\right)}{1-x}=\lim_{y\to 0}g(y)=\frac{\boldsymbol{\pi}}{\mathbf{2}}.$$

(d)
- $f(x)=-\frac{1}{x^2}\to-\infty$ as $x\to 0$ (a form $\frac{C}{0}$ with $C=-1<0$ and $x^2>0$), and $\lim_{y\to-\infty}e^y=0$. By case 3, $\lim_{x\to 0}e^{-1/x^2}=\mathbf{0}$.
- $f(x)=\frac{1}{x}\to+\infty$ as $x\to 0^+$ and $\lim_{y\to+\infty}e^y=+\infty$, so by case 3 $\lim_{x\to 0^+}e^{1/x}=\boldsymbol{+\infty}$. As $x\to 0^-$, $\frac{1}{x}\to-\infty$ and $\lim_{y\to-\infty}e^y=0$, so $\lim_{x\to 0^-}e^{1/x}=\mathbf{0}$. (In particular $e^{1/x}$ has no limit as $x\to 0$.)
- $f(x)=\sin x\to 0$ as $x\to 0^+$, and $\sin x>0$ for $0<x<\pi$. By the one-sided version of case 2 (see the remark after Theorem 6.12), $\lim_{x\to 0^+}\ln(\sin x)=\lim_{y\to 0^+}\ln y=\boldsymbol{-\infty}$.
- $f(x)=\frac{1}{x}\to\pm\infty$ as $x\to 0^{\pm}$, and $\lim_{y\to\pm\infty}\arctan y=\pm\frac{\pi}{2}$. By case 3, $\lim_{x\to 0^+}\arctan\frac{1}{x}=\frac{\boldsymbol{\pi}}{\mathbf{2}}$ and $\lim_{x\to 0^-}\arctan\frac{1}{x}=-\frac{\boldsymbol{\pi}}{\mathbf{2}}$ (a jump discontinuity at 0).

(3) We use Theorem 6.15 (the limit theorems for sequences) and the Ratio Test for sequences (Theorem 6.16).

(a) $a_n=\sqrt{n}\,(\sqrt{n}-1)$, and both factors tend to $+\infty$, so $a_n\to\boldsymbol{+\infty}$ (Theorem 6.15, item 6).

(b) $a_n>0$ and

$$\frac{a_{n+1}}{a_n}=\frac{(2n+2)!}{(n+1)!}\cdot\frac{n!}{(2n)!}=\frac{(2n+2)(2n+1)}{n+1}=2(2n+1)\xrightarrow{n\to\infty}+\infty>1,$$

so $a_n\to\boldsymbol{+\infty}$ by the Ratio Test.

(c) $a_n>0$ and

$$\frac{a_{n+1}}{a_n}=\frac{(2n+2)!}{((n+1)!)^2}\cdot\frac{(n!)^2}{(2n)!}=\frac{(2n+2)(2n+1)}{(n+1)^2}=\frac{2(2n+1)}{n+1}\xrightarrow{n\to\infty}4>1,$$

so $a_n\to\boldsymbol{+\infty}$ by the Ratio Test.

(d) Since $\binom{n}{3}=\frac{n(n-1)(n-2)}{6}$ (for $n\geq 3$; for $n=1,2$ both sides are 0),

$$a_n=\frac{n(n-1)(n-2)}{n^3}=\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\xrightarrow{n\to\infty}\mathbf{1}.$$

(e) Let $t_n=2^{-n}\pi$. Then $t_n\to 0$, $t_n\neq 0$, and $a_n=\pi\cdot\frac{\sin t_n}{t_n}$. By Theorem 6.15, item 7(b), with $g(t)=\frac{\sin t}{t}$ and Example 5.13, $a_n\to\pi\cdot 1=\boldsymbol{\pi}$.

(f) $\cos\left(\frac{n+1}{n}\cdot\frac{\pi}{2}\right)=\cos\left(\frac{\pi}{2}+\frac{\pi}{2n}\right)=-\sin\frac{\pi}{2n}$. With $t_n=\frac{\pi}{2n}\to 0$, $t_n\neq 0$:

$$a_n=-n\sin\frac{\pi}{2n}=-\frac{\pi}{2}\cdot\frac{\sin t_n}{t_n}\xrightarrow{n\to\infty}-\frac{\boldsymbol{\pi}}{\mathbf{2}},$$

again by Theorem 6.15, item 7(b), and Example 5.13.

(4) (a) Let $f(x)=\frac{x-\tan x}{x^2}$, defined for $x\neq 0$ near 0. Since tan is odd, $f(-x)=\frac{-x+\tan x}{x^2}=-f(x)$, so f is odd.

Right limit. For $0<x<\frac{\pi}{2}$ we have $\sin x<x<\tan x$, hence $\sin x-\tan x<x-\tan x<0$, and dividing by $x^2>0$,

$$\frac{\sin x-\tan x}{x^2}<f(x)<0.$$

For the lower bound,

$$\frac{\sin x - \tan x}{x^2} = \frac{\sin x\,(\cos x - 1)}{x^2 \cos x} = -\frac{\sin x}{x} \cdot \frac{1}{\cos x} \cdot x \cdot \frac{1 - \cos x}{x^2} \xrightarrow{x \to 0^+} -1 \cdot 1 \cdot 0 \cdot \frac{1}{2} = 0,$$

using Example 5.13 and $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$. By the Squeeze Theorem (Theorem 6.9, applied on the right of 0), $\lim_{x\to 0^+} f(x) = 0$.

Left limit. For $x < 0$ we have $f(x) = -f(-x)$, and $-x \to 0^+$, so $\lim_{x\to 0^-} f(x) = -\lim_{u\to 0^+} f(u) = 0$.

By Proposition 5.5, $\lim_{x\to 0} \frac{x - \tan x}{x^2} = \mathbf{0}$.

(b) Let $y = x - e$, so $y \to 0$ and $y \neq 0$ for $x \neq e$. Since $\ln x - 1 = \ln(e + y) - \ln e = \ln\left(1 + \frac{y}{e}\right)$,

$$\frac{\ln x - 1}{x - e} = \frac{\ln\left(1 + \frac{y}{e}\right)}{y} = \frac{1}{e} \cdot \frac{\ln(1+t)}{t}, \qquad t = \frac{y}{e}.$$

As $y \to 0$, $t \to 0$ with $t \neq 0$, so by case 2 of Theorem 6.12 and $\lim_{t\to 0} \frac{\ln(1+t)}{t} = 1$ (Section 6.7), the limit is $\boldsymbol{\frac{1}{e}}$.

(c) $\frac{2^{2x}-1}{2x} = \frac{1}{2} \cdot \frac{4^x - 1}{x} \to \frac{1}{2} \ln 4 = \mathbf{\ln 2}$, using $\lim_{x\to 0} \frac{a^x - 1}{x} = \ln a$ with $a = 4$ (Section 6.7); the right limit equals the limit.

(d) Let $y = x - 1 \to 0$, $y \neq 0$. Then $\ln x = \ln(1 + y)$ and $e^x - e = e\,(e^y - 1)$, so

$$\frac{\ln x}{e^x - e} = \frac{1}{e} \cdot \frac{\ln(1+y)}{y} \cdot \frac{y}{e^y - 1} \xrightarrow{y \to 0} \frac{1}{e} \cdot 1 \cdot 1 = \boldsymbol{\frac{1}{e}},$$

using $\frac{\ln(1+y)}{y} \to 1$ and $\frac{e^y - 1}{y} \to 1$ (Section 6.7) and the algebra of limits.

(e) For $x \neq 0$ near 0,

$$\frac{e^x - e^{-x}}{\sin x} = e^{-x} \cdot \frac{e^{2x} - 1}{\sin x} = e^{-x} \cdot \frac{e^{2x} - 1}{2x} \cdot 2 \cdot \frac{x}{\sin x} \xrightarrow{x \to 0} 1 \cdot 1 \cdot 2 \cdot 1 = \mathbf{2}.$$

Here $\frac{e^{2x}-1}{2x} \to 1$ by case 2 of Theorem 6.12 (with $t = 2x \to 0$, $t \neq 0$) and $\lim_{t\to 0} \frac{e^t - 1}{t} = 1$, and $\frac{x}{\sin x} \to 1$ by Example 5.13.

(f) The cube root and e^{-x^2} are defined for every real x, so $\mathrm{dom}(f) = \mathbb{R}$; the end-points of the domain are $\pm\infty$.

As $x \to +\infty$. For $x > 0$ we can write $\sqrt[3]{x} = e^{\frac{1}{3}\ln x}$, so $f(x) = e^{\frac{1}{3}\ln x - x^2}$. By the hint the exponent tends to $-\infty$, and $\lim_{t\to -\infty} e^t = 0$, so by case 3 of Theorem 6.12, $\lim_{x\to+\infty} f(x) = \mathbf{0}$. (This rewriting is only valid for $x > 0$, since $\ln x$ is not defined for $x \leq 0$.)

As $x \to -\infty$. f is odd: $f(-x) = \sqrt[3]{-x}\, e^{-x^2} = -f(x)$. Substituting $x = -u$ with $u \to +\infty$ (case 3),

$$\lim_{x\to-\infty} f(x) = \lim_{u\to+\infty} f(-u) = -\lim_{u\to+\infty} f(u) = \mathbf{0}.$$

(5) (a) (i) Let $a = 2^{-1/T} > 0$. Then $N(h) = N_0\, a^h$, and using $\lim_{h\to 0} \frac{a^h - 1}{h} = \ln a$ (Section 6.7),

$$\lim_{h\to 0^+} \frac{N(h) - N_0}{h} = N_0 \lim_{h\to 0^+} \frac{a^h - 1}{h} = N_0 \ln a = \boldsymbol{-\frac{N_0 \ln 2}{T}}.$$

The rate is negative (the number of atoms decreases), proportional to N_0, and inversely proportional to the half-life.

(ii) Let $y_n = \frac{1}{n}$, so $y_n \to 0$ and $y_n \neq 0$. Then $n\left(2^{1/n} - 1\right) = \frac{2^{y_n} - 1}{y_n}$, and by Theorem 6.15, item 7(b), with $g(y) = \frac{2^y - 1}{y}$ and $\lim_{y\to 0} g(y) = \ln 2$, the limit is $\mathbf{\ln 2}$.

(b) (i) $-\frac{t}{RC} \to -\infty$ as $t \to +\infty$ and $\lim_{s\to-\infty} e^s = 0$, so $e^{-t/(RC)} \to 0$ (case 3) and $\lim_{t\to+\infty} V(t) = \boldsymbol{V_0}$: the capacitor charges up to the voltage of the battery.

(ii) Let $s = \frac{t}{RC}$, so $s \to 0^+$, $s \neq 0$, as $t \to 0^+$. Then

$$\frac{V(t)}{t} = \frac{V_0}{RC} \cdot \frac{1 - e^{-s}}{s} = \frac{V_0}{RC} \cdot \frac{e^u - 1}{u}, \qquad u = -s,$$

and since $u \to 0$ with $u \neq 0$, case 2 and $\lim_{u\to 0} \frac{e^u - 1}{u} = 1$ give $\lim_{t\to 0^+} \frac{V(t)}{t} = \boldsymbol{\frac{V_0}{RC}}$. (The smaller the time constant RC, the faster the capacitor starts charging.)

(iii) $V(\tau) = V_0\left(1 - e^{-1}\right) \approx 0.632\, V_0$: after one time constant the capacitor has reached about **63%** of its final voltage.

(6) (a) (i) $\frac{\sin x/x}{x} = \sin x \cdot \frac{1}{x^2}$, where $\sin x$ is bounded and $\frac{1}{x^2} \to 0$ as $x \to \pm\infty$; by Corollary 6.10 this tends to 0, so $\frac{\sin x}{x} = o(x)$. Hence $\left(4 + \frac{\sin x}{x}\right) - 4 = o(x)$, which is what $4 + \frac{\sin x}{x} = 4 + o(x)$ means.

(ii) $\frac{\sin x}{x} = \sin x \cdot \frac{1}{x} \to 0$ (Corollary 6.10), so $\frac{\sin x}{x} = o(1)$. Moreover $\left|\frac{\sin x}{x}\right| \le 1 \cdot \left|\frac{1}{x}\right|$ for all $x \neq 0$, so $\frac{\sin x}{x} = \mathcal{O}\left(\frac{1}{x}\right)$ (with $C = 1$).

(iii) The statement with $\mathcal{O}\left(\frac{1}{x}\right)$ gives the most information: it says how fast $4 + \frac{\sin x}{x}$ approaches 4 (at least as fast as a constant times $\frac{1}{|x|}$). It implies the one with $o(1)$ (if $|f| \le \frac{C}{|x|}$ then $f \to 0$), which in turn implies the one with $o(x)$ (if $f \to 0$ then $\frac{f}{x} \to 0$). The statement $4 + o(x)$ does not even say that the function tends to 4: for example $f(x) = 4 + \sqrt{x}$ satisfies $\frac{\sqrt{x}}{x} = \frac{1}{\sqrt{x}} \to 0$, so $f(x) = 4 + o(x)$, but $f(x) \to +\infty$.

(b) We compute the limit of the quotient in each case, as $x \to +\infty$.

- $\frac{(x^2+3x)-x^2}{x^2} = \frac{3}{x} \to 0$: **true**.
- $\frac{x^2+3x}{x} = x + 3 \to +\infty$, so the quotient is not bounded on any neighborhood of $+\infty$: **false**.
- $\frac{\sqrt{x}}{x} = \frac{1}{\sqrt{x}} \to 0$: **true**.
- $\frac{x+\sin x}{x} = 1 + \frac{\sin x}{x} \to 1$: **true**.
- $(x + \sin x) - x = \sin x$ and $|\sin x| \le 1$: **true**.
- $\frac{x^2}{x^3} = \frac{1}{x} \to 0$: **true**.

(c) As $x \to 0$:

- $\frac{x^2+x^3}{x} = x + x^2 \to 0$: **true**.
- $\frac{x^3}{x^2} = x \to 0$: **true**.
- $\frac{x}{x^2} = \frac{1}{x}$, which is unbounded near 0: **false** (in fact $x^2 = o(x)$).
- $\frac{\sin x - x}{x} = \frac{\sin x}{x} - 1 \to 0$ (Example 5.13): **true**.
- $\frac{1-\cos x}{x} = x \cdot \frac{1-\cos x}{x^2} \to 0 \cdot \frac{1}{2} = 0$: **true**.
- $\frac{e^x - 1 - x}{x} = \frac{e^x - 1}{x} - 1 \to 0$ (Section 6.7): **true**.
- $\frac{\ln(1+x) - x}{x} = \frac{\ln(1+x)}{x} - 1 \to 0$ (Section 6.7): **true**.

(d) (i) $\frac{x^2+x}{x^2} = 1 + \frac{1}{x} \to 1$, so $x^2 + x \sim x^2$; but $(x^2 + x) - x^2 = x \to +\infty$.

(ii) In a formula, each $o(g)$ denotes *some* function which is $o(g)$, and two occurrences of $o(g)$ need not be the same function. In the example, $x^2 - 2x^2 = -x^2 \neq 0$. The correct rule is $o(g) - o(g) = o(g)$ (Proposition 8.1, item 1): here $-x^2 = o(x)$.

(e) Let $x = -\beta^2$; then $x \to 0$ and $x \neq 0$ as $\beta \to 0$, $\beta \neq 0$. With $\alpha = -\frac{1}{2}$, by case 2 of Theorem 6.12,

$$\frac{\gamma - 1}{\beta^2} = -\frac{(1+x)^{-1/2} - 1}{x} \xrightarrow{\beta \to 0} -\left(-\frac{1}{2}\right) = \frac{1}{2}.$$

Hence $\frac{\gamma - 1 - \frac{\beta^2}{2}}{\beta^2} \to 0$, i.e. $\gamma = 1 + \frac{\beta^2}{2} + o(\beta^2)$. Multiplying by mc^2 and using $\beta^2 = \frac{v^2}{c^2}$,

$$(\gamma - 1)mc^2 = \frac{1}{2}mv^2 + mc^2\, o\!\left(\frac{v^2}{c^2}\right) = \frac{1}{2}mv^2 + o(v^2) \qquad (v \to 0),$$

since multiplying by the constant mc^2, or replacing $\frac{v^2}{c^2}$ by v^2 inside $o(\cdot)$, does not change the class $o(\cdot)$ (Proposition 8.1, item 2, and the basic properties of Landau symbols).

(7) In this problem, "Cor." refers to Corollary 8.2 and "Prop." to Proposition 8.1. We also use that, as $x \to 0$, $o(x^n) \subseteq o(x^m)$ for $n \ge m$, i.e. every function which is $o(x^n)$ is also $o(x^m)$ (shown in the proof of Corollary 8.2), and that $x^n = o(x^m)$ for $n > m$ (comparison of powers).

(a) Expanding the product term by term:

$$\begin{aligned}&\left(x+3x^2+o(x^2)\right)(2-x+o(x))\\&=2x-x^2+\underbrace{x\,o(x)}_{o(x^2)\text{ (Cor.)}}+6x^2\underbrace{-3x^3}_{o(x^2)}+\underbrace{3x^2\,o(x)}_{o(x^3)\text{ (Cor., Prop. 2)}}+\underbrace{2\,o(x^2)}_{o(x^2)\text{ (Prop. 2)}}\underbrace{-x\,o(x^2)}_{o(x^3)\text{ (Cor.)}}+\underbrace{o(x^2)\,o(x)}_{o(x^3)\text{ (Cor.)}}\\&=2x+5x^2+o(x^2),\end{aligned}$$

since $o(x^3)\subseteq o(x^2)$ and a sum of finitely many $o(x^2)$ is $o(x^2)$ (Prop. 1). The coefficient of x^3 **cannot** be determined: the term $o(x^2)$ in the first factor could be, for instance, $c\,x^3$ for any $c\in\mathbb{R}$, and this changes the x^3 term of the product by $2c\,x^3$. The information about the factors only determines the product up to $o(x^2)$.

(b) Squaring,

$$(1+\alpha T+o(T))^2=1+2\alpha T+\underbrace{\alpha^2T^2}_{o(T)}+\underbrace{2\,o(T)}_{o(T)}+\underbrace{2\alpha T\,o(T)}_{o(T^2)\subseteq o(T)}+\underbrace{o(T)\,o(T)}_{o(T^2)\subseteq o(T)}=1+2\alpha T+o(T),$$

so $A(T)=L(T)^2=L_0^2\,(1+2\alpha T+o(T))$. Then

$$\begin{aligned}V(T)=L(T)\,A(T)&=L_0^3\,(1+\alpha T+o(T))\,(1+2\alpha T+o(T))\\&=L_0^3\left(1+3\alpha T+\underbrace{2\alpha^2T^2}_{o(T)}+o(T)\right)=L_0^3\,(1+3\alpha T+o(T))\,,\end{aligned}$$

where all the products involving $o(T)$ are $o(T)$ by the same rules. Thus $\frac{A(T)-L_0^2}{L_0^2}=2\alpha T+o(T)$ and $\frac{V(T)-L_0^3}{L_0^3}=3\alpha T+o(T)$, while $\frac{L(T)-L_0}{L_0}=\alpha T+o(T)$.

(c)

$$(1+2t+o(t))\,(1-t+o(t))=1-t+2t\underbrace{-2t^2}_{o(t)}+\underbrace{o(t)+2t\,o(t)+o(t)-t\,o(t)+o(t)\,o(t)}_{o(t)}=1+t+o(t),$$

so $P(t)=V_0I_0\,(1+t+o(t))$. The relative change of the power, t, is (to first order) the sum $2t+(-t)$ of the relative changes of the voltage and the current.

(d) Expanding:

$$(2+o(1))\left(x-x^3+o(x^3)\right)=2x\underbrace{-2x^3}_{o(x)}+\underbrace{2\,o(x^3)}_{o(x^3)\subseteq o(x)}+\underbrace{x\,o(1)}_{o(x)\text{ (Prop. 3)}}\underbrace{-x^3\,o(1)}_{o(x^3)\subseteq o(x)}+\underbrace{o(1)\,o(x^3)}_{o(x^3)\subseteq o(x)}=2x+o(x).$$

The term $x\,o(1)$ is only known to be $o(x)$, and it can be much larger than x^3. For instance, $x=o(1)$ as $x\to0$, and $(2+x)(x-x^3)=2x+x^2-2x^3-x^4$, which is *not* of the form $2x-2x^3+o(x^3)$, because x^2-x^4 is not $o(x^3)$.

(e) The difference is $o(x^2)-o(x^3)$. Since $o(x^3)\subseteq o(x^2)$, this is $o(x^2)-o(x^2)=o(x^2)$ (Prop. 1). It is not 0, because the two o's denote possibly different functions: for example, $x+2x^2+x^3$ is of the form $x+2x^2+o(x^2)$ and $x+2x^2$ is of the form $x+2x^2+o(x^3)$, and their difference is $x^3\neq0$.

(f) As $x\to+\infty$, every constant is $o(x)$ (since $\frac{c}{x}\to0$), and $o(1)\subseteq o(x)$ (if $\frac{f}{1}\to0$ then $\frac{f}{x}=f\cdot\frac{1}{x}\to0$). Expanding,

$$\begin{aligned}&\left(x^2+x+o(x)\right)\left(1+\frac{1}{x}+o\left(\frac{1}{x}\right)\right)\\&=x^2+x+\underbrace{x^2\,o\left(\tfrac{1}{x}\right)}_{o(x)\text{ (Prop. 3)}}+x+\underbrace{1}_{o(x)}+\underbrace{x\,o\left(\tfrac{1}{x}\right)}_{o(1)\text{ (Prop. 3)}}+\underbrace{o(x)}_{}+\underbrace{\tfrac{1}{x}\,o(x)}_{o(1)\text{ (Prop. 3)}}+\underbrace{o(x)\,o\left(\tfrac{1}{x}\right)}_{o(1)\text{ (Prop. 4)}}\\&=x^2+2x+o(x).\end{aligned}$$

Bonus problems.

(B1) (a) *Theorem 6.16 (Ratio Test).* Let $\{a_n\}_{n\in\mathbb{N}}$ be positive for all large n, and suppose that $\lim_{n\to\infty}\frac{a_{n+1}}{a_n} = q$ exists (finite or infinite). If $q<1$, then $\lim_{n\to\infty} a_n = 0$; if $q>1$, then $\lim_{n\to\infty} a_n = +\infty$; if $q=1$, the test is inconclusive.

(b) For all large n we have $\frac{a_{n+1}}{a_n} > 0$, so by the comparison of limits (Theorem 6.15, item 4, comparing with the constant sequence 0) $q \geq 0$. Since $0 \leq q < 1$, we can take for instance $r = \frac{q+1}{2}$, so that $q < r < 1$.

(c) Let $\varepsilon = r - q > 0$. By the definition of the limit there is $N_1 \in \mathbb{N}$ with $\left|\frac{a_{n+1}}{a_n} - q\right| < \varepsilon$, hence $\frac{a_{n+1}}{a_n} < q + \varepsilon = r$, for all $n > N_1$. By assumption there is N_2 with $a_n > 0$ for all $n > N_2$. Let $N = \max\{N_1, N_2\}$. For $n > N$ we have $a_n > 0$, and multiplying by $a_n > 0$ gives $a_{n+1} < r\, a_n$.

(d) *Proof.* For $k = 0$ the claim reads $0 < a_M \leq a_M$, which holds since $M > N$. Suppose $0 < a_{M+k} \leq r^k a_M$. Since $M + k > N$, by (c) $a_{M+k+1} < r\, a_{M+k} \leq r^{k+1} a_M$, and $a_{M+k+1} > 0$ because $M + k + 1 > N$. This proves the claim for $k+1$. □

(e) For $n \geq M$, writing $n = M + k$, (d) gives $0 < a_n \leq a_M\, r^{n-M} = \left(a_M r^{-M}\right) r^n$. Since $r^n \to 0$, the right-hand side tends to 0, and by the Squeeze Theorem for sequences (Theorem 6.15, item 5(a)) $\lim_{n\to\infty} a_n = 0$.

(f) If $q \in \mathbb{R}$, $q > 1$, let $r = \frac{1+q}{2}$ and $\varepsilon = q - r > 0$: as in (c) there is N such that $a_n > 0$ and $\frac{a_{n+1}}{a_n} > q - \varepsilon = r$ for $n > N$. If $q = +\infty$, the definition of the limit (with $A = 2$) gives N such that $a_n > 0$ and $\frac{a_{n+1}}{a_n} > 2 = r$ for $n > N$. In both cases $a_{n+1} > r\, a_n$ for $n > N$, and the same induction as in (d) gives $a_{M+k} \geq r^k a_M$, i.e. $a_n \geq \left(a_M r^{-M}\right) r^n$ for $n \geq M$. Since $a_M r^{-M} > 0$ and $r^n \to +\infty$, the right-hand side tends to $+\infty$, and by the Squeeze to $\pm\infty$ Theorem for sequences (Theorem 6.15, item 5(b)) $\lim_{n\to\infty} a_n = +\infty$.

(g) $a_n = \frac{1}{n}$: $\frac{a_{n+1}}{a_n} = \frac{n}{n+1} \to 1$ and $a_n \to 0$. $a_n = 1$: the ratio is 1 and $a_n \to 1$. $a_n = n$: $\frac{a_{n+1}}{a_n} = \frac{n+1}{n} \to 1$ and $a_n \to +\infty$. Since the same value $q = 1$ occurs for sequences with different limits, knowing that $q = 1$ does not determine the limit.

(h) For $a_n = \frac{n^{10}}{1.1^n}$: $\frac{a_{n+1}}{a_n} = \frac{1}{1.1}\left(1 + \frac{1}{n}\right)^{10} \to \frac{1}{1.1} < 1$, so $a_n \to \mathbf{0}$. For $a_n = \frac{100^n}{n!}$: $\frac{a_{n+1}}{a_n} = \frac{100}{n+1} \to 0 < 1$, so $a_n \to \mathbf{0}$.

(B2) (a) (i) If $f \sim g$, then $\lim_{x\to x_0}\frac{f}{g} = 1$, which is finite and nonzero; by definition, $f \asymp g$.

(ii) Here $\ell \in \mathbb{R}$ and $\ell \neq 0$, so $\ell g \neq 0$ on a deleted neighborhood of x_0, and $\frac{f}{\ell g} = \frac{1}{\ell}\cdot\frac{f}{g} \to \frac{1}{\ell}\cdot\ell = 1$. Hence $f \sim \ell g$.

(iii) As $x \to +\infty$, let $f(x) = x$ and $g(x) = x^2$. For $x \geq 1$ we have $|f(x)| \leq |g(x)|$, so $f = \mathcal{O}(g)$. But $\frac{f}{g} = \frac{1}{x} \to 0$, which is not a nonzero finite number, so $f \not\asymp g$ (in fact $f = o(g)$).

(b) *Proof.* On a deleted neighborhood of x_0 we have $\frac{f-g}{g} = \frac{f}{g} - 1$. By the algebra of limits, $\frac{f}{g} \to 1$ if and only if $\frac{f}{g} - 1 \to 0$, i.e. if and only if $\frac{f-g}{g} \to 0$. So $f \sim g$ if and only if $f - g = o(g)$, which is what $f = g + o(g)$ means. □

(c) (i) *Proof.* Since $\frac{f}{g} \to 1 > 0$, by Theorem 6.2 (local sign) there is a deleted neighborhood of x_0 on which $\frac{f}{g} > 0$; in particular $f \neq 0$ there. On this neighborhood $\frac{g}{f} = \frac{1}{f/g}$, and by the algebra of limits $\frac{g}{f} \to \frac{1}{1} = 1$. Hence $g \sim f$. □

(ii) *Proof.* On a deleted neighborhood of x_0 where $g \neq 0$ and $h \neq 0$ (the intersection of two deleted neighborhoods), $\frac{f}{h} = \frac{f}{g}\cdot\frac{g}{h} \to 1 \cdot 1 = 1$. Hence $f \sim h$. □

(iii) *Proof.* Since $g = \mathcal{O}(h)$, there are $C > 0$ and a deleted neighborhood of x_0 on which $|g| \leq C|h|$; shrinking it, we may assume that also $g \neq 0$ and $h \neq 0$ there. Then $\left|\frac{g}{h}\right| \leq C$, i.e. $\frac{g}{h}$ is bounded, while $\frac{f}{g} \to 0$ because $f = o(g)$. By Corollary 6.10,

$$\frac{f}{h} = \frac{f}{g}\cdot\frac{g}{h} \to 0,$$

so $f = o(h)$. (Note that $\frac{g}{h}$ need not have a limit, so we cannot simply use the algebra of limits.) □

(d) *Proof.* By definition (with $g = 1$), $f(x) = f(x_0) + o(1)$ means $f(x) - f(x_0) = o(1)$, i.e.

$$\lim_{x \to x_0} (f(x) - f(x_0)) = 0,$$

which by the algebra of limits is equivalent to $\lim_{x \to x_0} f(x) = f(x_0)$, i.e. to the continuity of f at x_0. □

Problems for later.

(L1) (a) (i) *Proof.* $(1-r)s_n = \sum_{k=0}^{n} r^k - \sum_{k=0}^{n} r^{k+1} = (1 + r + \cdots + r^n) - \left(r + r^2 + \cdots + r^{n+1}\right) = 1 - r^{n+1}$, since all the other terms cancel. Dividing by $1 - r \neq 0$ gives the formula. □

(ii) $2.3\overline{17} = \frac{23}{10} + \frac{17}{10^3} + \frac{17}{10^5} + \cdots = \frac{23}{10} + \frac{17}{1000} \sum_{j=0}^{\infty} \left(\frac{1}{100}\right)^j = \frac{23}{10} + \frac{17}{1000} \cdot \frac{1}{1 - \frac{1}{100}} = \frac{23}{10} + \frac{17}{990} = \frac{2294}{990} = \mathbf{\frac{1147}{495}}$.

(iii) After the n-th bounce the ball rises to the height $10 \cdot (0.6)^n$ m, and then falls the same distance. The total distance is a geometric series with ratio $0.6 < 1$:

$$10 + 2\sum_{n=1}^{\infty} 10 \cdot (0.6)^n = 10 + 20 \cdot \frac{0.6}{1 - 0.6} = 10 + 30 = \mathbf{40} \text{ m}.$$

The first fall takes $\sqrt{2 \cdot 10/g}$ seconds, and the n-th bounce (up and down) takes

$$2\sqrt{\frac{2 \cdot 10 \cdot (0.6)^n}{g}} = 2\sqrt{\frac{20}{g}} \left(\sqrt{0.6}\right)^n \text{ seconds}.$$

The total time is

$$\sqrt{\frac{20}{g}} \left(1 + 2\sum_{n=1}^{\infty} \left(\sqrt{0.6}\right)^n\right) = \sqrt{\frac{20}{g}} \left(1 + \frac{2\sqrt{0.6}}{1 - \sqrt{0.6}}\right) \approx 1.428 \cdot 7.873 \approx \mathbf{11.2} \text{ s}.$$

This is a convergent geometric series (ratio $\sqrt{0.6} < 1$), so the infinitely many bounces take a finite time, after which the ball rests on the ground — just like Achilles catching up with the tortoise.

(iv) Just after the n-th dose, the body contains the new 100 mg, 25 mg left from the previous dose, $100 \cdot (0.25)^2$ mg from the one before, and so on:

$$A_n = 100 \sum_{k=0}^{n-1} (0.25)^k = 100 \cdot \frac{1 - (0.25)^n}{1 - 0.25} = \frac{400}{3}\left(1 - 4^{-n}\right) \text{ mg} \xrightarrow{n \to \infty} \mathbf{\frac{400}{3} \approx 133.3} \text{ mg}.$$

(b) Let $a_n = \frac{1}{n(\ln n)^q}$ for $n \geq 2$. Since $\ln n > 0$ for $n \geq 2$, $a_n > 0$; and $n(\ln n)^q$ is a product of two positive increasing functions of n (recall $q > 0$), so it is increasing and a_n is decreasing. Theorem 7.1 is stated for series starting at $n = 1$; setting $a_1 = a_2$ we obtain a non-negative decreasing sequence $\{a_n\}_{n \in \mathbb{N}_+}$, and adding the single term a_1 changes every partial sum by the same constant, so it does not affect convergence. The condensed series has terms ($k \geq 1$)

$$2^k a_{2^k} = \frac{2^k}{2^k (\ln 2^k)^q} = \frac{1}{(k \ln 2)^q} = \frac{1}{(\ln 2)^q} \cdot \frac{1}{k^q}.$$

Up to the positive constant factor $(\ln 2)^{-q}$ (which does not affect convergence), this is the p-series with $p = q$, which converges if and only if $q > 1$ (Corollary 7.2). By the Cauchy Condensation Test, $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^q}$ **converges if and only if** $q > 1$. (For $q = 1$ this is a series that diverges even more slowly than the harmonic series.)

(c) Let $b_k = \sqrt{\frac{k^3+3}{2k^3-5}}$, defined for $k \geq 2$ (where $2k^3 - 5 > 0$). Since $\frac{k^3+3}{2k^3-5} = \frac{1+3/k^3}{2-5/k^3} \to \frac{1}{2}$ and the square root is continuous, $b_k \to \frac{1}{\sqrt{2}}$. The terms of the series are $a_k = (-1)^k b_k$, so $a_{2j} \to \frac{1}{\sqrt{2}}$ and $a_{2j+1} \to -\frac{1}{\sqrt{2}}$. If $a_k \to 0$, every subsequence would also tend to 0; hence $a_k \not\to 0$, and by the necessary condition the series **does not converge**. Leibniz's Test cannot be applied because its third hypothesis, $b_k \to 0$, fails.

(d) (i) $a_k = \frac{1}{2^k-1} > 0$, and with $b_k = \left(\frac{1}{2}\right)^k$ we have $\frac{a_k}{b_k} = \frac{2^k}{2^k-1} = \frac{1}{1-2^{-k}} \to 1$. Since the geometric series $\sum \left(\frac{1}{2}\right)^k$ converges, the series **converges** by the Asymptotic Comparison Test. (Alternatively, $2^k - 1 \geq 2^{k-1}$ gives $a_k \leq 2\left(\frac{1}{2}\right)^k$, and we can use the Comparison Test.)

(ii) With $b_k = \frac{1}{k^2}$:

$$\frac{a_k}{b_k} = \frac{k^2(k+3)}{\sqrt[3]{k^9+k^2}} = \frac{k^3\left(1+\frac{3}{k}\right)}{k^3\sqrt[3]{1+k^{-7}}} = \frac{1+\frac{3}{k}}{\sqrt[3]{1+k^{-7}}} \to 1.$$

Since $\sum \frac{1}{k^2}$ converges (p-series, $p=2$), the series **converges** by the Asymptotic Comparison Test.

(e) All terms are positive.

(i) $\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1+\frac{1}{n}\right)^n} \to \frac{1}{e} < 1$: **converges** by the Ratio Test.

(ii) The ratio is the reciprocal of the one in (i), so it tends to $e > 1$: **diverges** by the Ratio Test.

(iii) The ratios are $2\left(\frac{n}{n+1}\right)^n \to \frac{2}{e} < 1$ and $3\left(\frac{n}{n+1}\right)^n \to \frac{3}{e} > 1$ (recall $2 < e < 3$): the first series **converges** and the second **diverges**, by the Ratio Test.

(iv) $\frac{a_{n+1}}{a_n} = \frac{1}{3}\left(\frac{n+1}{n}\right)^3 \to \frac{1}{3} < 1$: **converges** by the Ratio Test.

(v) $\sqrt[n]{a_n} = \frac{n}{2n+1} \to \frac{1}{2} < 1$: **converges** by the Root Test.

(f) (i) By the hint, $\sin\left(k\pi + \frac{1}{k}\right) = (-1)^k \sin\frac{1}{k}$. Let $b_k = \sin\frac{1}{k}$. Since $0 < \frac{1}{k} \leq 1 < \frac{\pi}{2}$, we have $b_k > 0$; since $\frac{1}{k}$ is decreasing and sin is increasing on $\left[0, \frac{\pi}{2}\right]$, b_k is decreasing; and $b_k \to \sin 0 = 0$ by continuity. By Leibniz's Test the series converges. It does not converge absolutely, since $\sum \sin\frac{1}{k}$ diverges (an example in Section 7.5 of the lecture notes). So it **converges conditionally**.

(ii) Let $b_k = \frac{k^2}{k^3+1}$. Then $b_k > 0$ and $b_k \to 0$. Moreover b_k is decreasing: $b_k \geq b_{k+1}$ is equivalent to $k^2\left((k+1)^3+1\right) \geq (k+1)^2\left(k^3+1\right)$, i.e. (expanding both sides)

$$k^4 + 2k^3 + k^2 - 2k - 1 \geq 0,$$

which holds for every $k \geq 1$ because $k^4 - 1 \geq 0$, $2k^3 - 2k \geq 0$ and $k^2 > 0$. By Leibniz's Test (the overall sign $(-1)^{k+1} = -(-1)^k$ does not matter) the series converges. Since $\frac{b_k}{1/k} = \frac{k^3}{k^3+1} \to 1$ and the harmonic series diverges, $\sum b_k$ diverges by the Asymptotic Comparison Test, so the series **converges conditionally**.

(iii) Let $b_k = \ln\left(1+\frac{1}{k}\right)$. Since $1 + \frac{1}{k} > 1$ is decreasing and ln is increasing, $b_k > 0$ is decreasing, and $b_k \to \ln 1 = 0$. By Leibniz's Test the series converges. Moreover, with $t_k = \frac{1}{k} \to 0$, $t_k \neq 0$, we have $\frac{b_k}{1/k} = \frac{\ln(1+t_k)}{t_k} \to 1$ (Theorem 6.15, item 7(b)), so $\sum b_k$ diverges by comparison with the harmonic series, and the series **converges conditionally**.

(g) (i) Let $c_k = |a_k| = \frac{5^{k-1}}{(k+1)^2\,4^{k+2}}$. Then

$$\frac{c_{k+1}}{c_k} = \frac{5}{4}\left(\frac{k+1}{k+2}\right)^2 \to \frac{5}{4} > 1,$$

so $\sum c_k$ diverges by the Ratio Test: the series does **not** converge absolutely. Moreover, by the Ratio Test for sequences (Theorem 6.16), $c_k \to +\infty$, so $a_k \not\to 0$ and the series **does not converge** either (see the remark after the Absolute Convergence Test).

(ii) $\left|\frac{\cos 3k}{k^3}\right| \leq \frac{1}{k^3}$ and $\sum \frac{1}{k^3}$ converges (p-series, $p=3$), so by the Comparison Test the series **converges absolutely** (and therefore converges).

(L2) (a) (i) Each of the 2^{k-1} numbers $\frac{1}{2^{k-1}+1}, \ldots, \frac{1}{2^k}$ is at least $\frac{1}{2^k}$, so their sum is at least $2^{k-1} \cdot \frac{1}{2^k} = \frac{1}{2}$.

(ii) Grouping the terms of H_{2^m} into the first term and m groups as in (i),

$$H_{2^m} = 1 + \sum_{k=1}^{m}\left(\frac{1}{2^{k-1}+1} + \cdots + \frac{1}{2^k}\right) \geq 1 + m \cdot \frac{1}{2}.$$

(For $m = 0$ this reads $H_1 = 1 \geq 1$.)

(iii) H_n is increasing, since all terms are positive. Given $A > 0$, choose $m \in \mathbb{N}$ with $1 + \frac{m}{2} > A$. Then for every $n > 2^m$ we have $H_n \geq H_{2^m} \geq 1 + \frac{m}{2} > A$. Hence $\lim_{n\to\infty} H_n = +\infty$.

(b) By (a)(ii), $H_{2^m} \geq 10$ as soon as $1 + \frac{m}{2} \geq 10$, i.e. $m \geq 18$. So $n = 2^{18} = \mathbf{262144}$ works. (In fact the smallest such n is 12367: the harmonic series diverges, but very slowly.)

(c) (i) The center of mass of book 1 is at its midpoint, at distance 1 from its right edge. Book 1 rests on book 2 and does not tip over as long as its center of mass is not beyond the right edge of book 2, i.e. as long as $d_1 \leq 1$. For $d_1 = 1$ its center of mass lies exactly above the right edge of book 2.

(ii) The right edge of book $k+1$ is at the origin and the book has length 2, so its center of mass (its midpoint) is at $x = -1$.

(iii) The top k books are identical, and the average of their centers of mass is 0 by assumption, so the sum of their positions is $k \cdot 0 = 0$. Adding book $k+1$, the center of mass of the top $k+1$ books is

$$\frac{0 + (-1)}{k+1} = -\frac{1}{k+1}.$$

(iv) The top $k+1$ books rest on book $k+2$ (or on the table, if $k+1 = n$), whose right edge is at $x = -d_{k+1}$. They do not tip over as long as their center of mass is not beyond this edge, i.e. as long as $-\frac{1}{k+1} \leq -d_{k+1}$, i.e. $d_{k+1} \leq \frac{1}{k+1}$. For $d_{k+1} = \frac{1}{k+1}$ the center of mass lies exactly above the right edge of book $k+2$ (or of the table).

(v) Let $P(k)$ be the statement: if $d_j = \frac{1}{j}$ for $j = 1, \ldots, k$, then the center of mass of the top k books lies exactly above the right edge of book $k+1$ (or of the table, if $k = n$). $P(1)$ holds by (i), and $P(k) \Rightarrow P(k+1)$ by (ii)–(iv). So $P(k)$ holds for all $k = 1, \ldots, n$, and the stack with $d_k = \frac{1}{k}$ does not tip over. The right edge of the top book is then at distance $d_1 + d_2 + \cdots + d_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} = H_n$ beyond the edge of the table.

(vi) We need $H_n > 3$. By (a)(ii), $H_{2^m} \geq 1 + \frac{m}{2} > 3$ as soon as $m \geq 5$, so $n = 2^5 = \mathbf{32}$ books are enough. With a calculator: $H_{10} \approx 2.929 < 3$ and $H_{11} \approx 3.020 > 3$, so 11 **books suffice, but** 10 **do not**. (Since H_{11} is strictly larger than 3, we can choose the d_k slightly smaller than $\frac{1}{k}$, so that the stack is not exactly at the tipping point, and still have an overhang larger than 3.)

(vii) **Yes.** Given any distance $D > 0$, since $H_n \to +\infty$ by (a), there is n with $H_n > D$, and a stack of n books reaches beyond D (again choosing the d_k slightly smaller than $\frac{1}{k}$).